Lab 03: Compiler optimization

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Compile with optimization option

GCC and its variations provide option for optimizing code

- -O0: Generates code with no optimization. Recommend for debugging, as the generated code closely matches the original source code. (Default)
- -O1: Enables basic optimizations that do not require a great deal of compile time (i.e. inlining of functions, constant propagation, and dead code elimination).
- -O2: Includes -O1 and optimizations such as loop unrolling and better register allocation. Slower compiling time compared to -O1
- -O3: Includes -O2 and loop optimizations, vectorization, and inlining. Might lead to longer compilation times and potentially larger binary sizes.
- -Ofast: Beyond -O3 and sacrifice some IEEE or C/C++ standard compliance in favor of speed. Including invalid optimization for all programs, such as allowing unsafe math optimizations.

Example:

```
g++ -O3 -o main.cpp main
```

Practice

Run the program in the next slide with different compiler optimization mode

- Record run time (For each optimization mode, you should run at least 10 times and get the median)
- Compare run time between optimization mode
- What does each optimization mode do when compile this program? Please show details, which part of the code is optimized by which method. (This is an open question)

Practice

```
#include <iostream>
#include <chrono>
int main() {
    const int N = 100000000; //
Number of iterations
    double sum = 0.0;
    // Start timing
    auto start =
std::chrono::high_resolution_clock
::now();
    // Perform a computationally
intensive task
    for (int i = 0; i < N; ++i) {
        sum += 1.0 / (i + 1);
    }
    // Stop timing
    auto end =
std::chrono::high_resolution_clock
::now();
    std::chrono::duration<double>
elapsed = end - start;
    // Output the result and the
time taken
    std::cout << "Sum: " << sum <<
std::endl;
    std::cout << "Time taken: " <<
elapsed.count() << " seconds" <<
std::endl;
}
```

Submission

A PDF file containing your work on Google Colab